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Valentina Vecchio

Research Associate

Links:
Personal Website
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I am an experienced researcher with a passion for discovering statistical insights within large data. Currently, I am a convener for the ATLAS collaboration at CERN, an international scientific experiment that received the 2025 Breakthrough Prize and contributed to the discovery of the Higgs boson, which led to the 2013 Nobel Prize. The aim of the collaboration is to advance the field of particle physics through the statistical analysis of a large dataset. During my 10 years in research I have developed a wide range of data-science skills:

- data manipulation and visualisation,
- statistical modelling,
- usage of machine-learning techniques for classification and regression problems,
- scientific writing and communication to both expert and non-expert audiences.

I have covered several leadership roles and mentored numerous Ph.D students. I am excited to leverage my strong analytical skills, statistical modelling and machine learning expertise to solve real-world data challenges.

Technical Skills

Tools & Languages	Python, C++, Git, Bash, LaTeX, MarkDown
	Polars, Pandas, NumPy, matplotlib, boost-histogram, ROOT
Quantitative Research	Mathematical optimization, Mathematical Modelling, Machine Learning
Communication	English (fluent), Italian (native)

Leadership & Experience

The ATLAS experiment is a large international collaboration of roughly 4k scientists. The collaboration is organised in groups covering the different topics of the physics programme. Every group has an official mandate and it is lead by two group conveners.

Within the ATLAS collaboration I have covered several leadership roles, learning how to guide a team into exploring new data-analysis techniques while ensuring the robustness needed to produce scientific results.

Flavour Tagging Convener / ATLAS Collaboration **October 2024 — Present**
University of Manchester

I was elected group convener by the executive board of the experiment. As convener I directed the group activities, organised topical workshops and publications, and arranged interdisciplinary collaboration.

I lead a dedicated group of 200 researchers and Ph.D students organised in four sub-groups (2 of which I convened in the past). This team uses cutting-edge machine learning to advance the field of particle physics while ensuring robustness of the models deployed through data-driven calibrations.

Every sub-group is lead by two conveners and cover the following topics:

- algorithms: development and validation of cutting-edge machine learning algorithms (from simple Neural Networks, to DeepSets and Transformer models) using Monte Carlo simulated data for the identification of the flavour of jets.
- calibration: measurement of the performance of the algorithms in real data and comparison with expected performance
- software: deployment of the taggers and the associated data-to-simulation correction in the main software used by the members of the experiment
- $X \to b\bar{b}$: development and calibration of algorithms for the identification of boosted Higgs boson.

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I provide strategic guidance to the group to meet the collaboration's broader goal of driving innovative research and discovery. My work consists of liaising between senior stakeholders and members of my team to translate strategic research priorities into achievable deliverables. I review the work in each sub-group and help the teams to define measurable benchmarks to assess analytical effectiveness. Finally, I am the point of contact for management and any researcher of the collaboration who needs information or support in using the group's work.

Flavour Tagging Calibration Convener / ATLAS Collaboration — **October 2022 — October 2024**
University of Manchester

I lead the team that provided the first ever data-driven calibration of a transformer tagger for the identification of hadronic jets (ask me what that means!). I have coordinated different calibration activities, providing feedback on the results produced and identifying points of possible improvements or bugs. I developed several plotting tools to validate the result of the calibrations and improved the documentation provided to the rest of the collaboration.

Flavour Tagging boosted $X \to bb$ contact / ATLAS Collaboration — **October 2020 — October 2022**
University of Manchester

I lead the team that provided the first ever data-driven calibration of a Neural Network tagger for the identification highly energetic Higgs boson objects. I conducted validation studies on the tagger, to understand its performance on Monte Carlo simulated events. Moreover, I designed a brand new data-driven method for the measurement of the tagger performance and developed a code-base that allows the full chain of calibration to run with less steps and in less time. I pioneered the development of a simple Boosted Decision Tree algorithm to increase the purity of a very challenging calibration data-set.

Analysis Contact of Higgs boson analysis / ATLAS Collaboration — **March 2021 — Present**
University of Manchester

I designed a complex analysis for the first-ever estimation of the sign of the top-quark Yukawa coupling to the Higgs boson. The analysis consisted of several statistical orthogonal channels separately optimised to increase their statistical power. Boosted Decision trees were used to remove the overwhelmingly high background. I designed a profile-likelihood fit for the statistical inference analysis.

Flavour Tagging at Muon Colliders / UK Muon Collider — **March 2024 — Present**
University of Manchester

Muon Colliders are theorised to be the future of particle physics. They can achieve high energies in a more compact accelerator complex and will open the door for exciting research never attempted before by humankind. In 2024 I made contact with other UK researchers interested in demonstrating the physics reach of this kind of collider. Together with my UK colleagues, we setup a team for the development of a transformer-based flavour tagging algorithm which will allow for the precise measurement of the properties of the Higgs boson.

Education

PhD in Physics, *Università degli Studi Roma Tre* — March 2020
Master Degree in Physics, *Università degli Studi Roma Tre* — October 2017
Bachelor Degree in Physics, *Università degli Studi Roma Tre* — September 2014